Ali Elouafiq
SQLI Digital Lab, Rabat, Morocco
aelouafiq@sqli.com

14 June 2026

Dear Annalisa Crannell,

I would like to submit my article *The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity* for consideration by *The American Mathematical Monthly*. The work is original, unpublished, and not under consideration elsewhere.

Here is the observation at its heart. Analysts have spent a century carefully cataloguing the regularity conditions—continuity, measurability, monotonicity, boundedness—that force a solution of Jensen's equation to be affine. Every one of those conditions becomes superfluous the moment the mixing weight p ranges over the whole interval $[0, 1]$ rather than just the rationals. A single substitution then reads the answer directly off the chord, with no hypothesis on the function at all. The proof is one line; the surprise is what it implies.

What it implies is a clean divide running through a surprising range of mathematics. In expected-utility theory and the axiomatic characterization of Shannon entropy, the classical regularity hypotheses are doing genuine, irreplaceable work. In calibration theory on an atomless probability space, in Gleason-type derivations of the quantum Born rule on the preparation side, in integral representations of proper scoring rules—they are dead weight, carried along by habit. The paper makes the divide explicit: a small dictionary, a structural account of why Hamel's pathological solutions survive the rational weights but not the irrational ones, and a short diagnostic a reader can apply to their own derivation.

I think *Monthly* readers will enjoy this. The result itself is accessible to anyone who has seen a functional equation; the territory it maps is broad enough to reward a much wider audience; and there is a genuine puzzle at the centre—why does one extra weight change everything?—of the kind the *Monthly* does best.

I have no conflicts of interest to disclose.

Please address correspondence to me at aelouafiq@sqli.com. Thank you for your consideration.

Sincerely,

Ali Elouafiq